

ACYUTH GOPALAKRISHNAN

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EDUCATION

CARNEGIE MELLON UNIVERSITY TEPPER SCHOOL OF BUSINESS
Master of Science in Business Analytics - MSBA

Pittsburgh, PA
05/27

VELLORE INSTITUTE OF TECHNOLOGY
Bachelors of Technology in Computer Science and Engineering
Specialization: Data Science

Vellore, Tamil Nadu
05/26

- Relevant Coursework: Machine Learning, Deep Learning, Predictive Analytics, Database Management Systems, Statistics, Artificial Intelligence, Data Structures and Algorithms

PROJECT WORK EXPERIENCE

VELLORE INSTITUTE OF TECHNOLOGY
TOLL TRAFFIC BEHAVIOUR AND STRESS ASSESSMENT USING
VISION BASED ANALYSIS (Capstone)

12/25 – 05/26

- **Computer Vision & AI:** Engineered a dual stream deep learning framework integrating YOLOv8 for traffic extraction and a VGG16 model to correlate congestion with driver stress, with the end goal of fusing vehicle kinematics and behavioural biometrics into a unified safety index to mitigate post congestion road accidents.
- **Data Evaluation:** Benchmarked 10 distinct architectures under a strict subject disjoint protocol on a 22,000+ image dataset to isolate the optimal accuracy efficiency frontier; achieved a peak 94.58% validation accuracy and established the exact baseline parameters required for edge deployment on live toll infrastructure.

VELLORE INSTITUTE OF TECHNOLOGY (NIMHANS Collaboration)
DEEP LEARNING METHODS FOR TRAUMATIC BRAIN INJURY

06/25 – 12/25

- **Predictive Modeling:** Developed a deep learning model utilizing a Vision Transformer with Triplet Attention to predict TBI outcomes from CT scans, optimizing diagnostic accuracy to a 0.91 AUC-ROC across a complex multi label classification framework.
- **Cross-Functional Leadership:** Directed a 3-person technical team under faculty supervision to validate model performance via stratified sampling, generating interpretable attention heatmaps to enhance clinical decision support.

PROFESSIONAL WORK EXPERIENCE

HDFC BANK
Credit Risk Analytics and Innovation Intern

Chennai, Tamil Nadu
05/25 – 07/25

- **Data Analysis:** Built scalable data wrangling pipelines using Pandas and Excel to analyze complex transaction behaviours, resulting in a **30%** improvement in feature engineering efficiency for the credit risk team.
- **Problem Solving:** Applied statistical modeling to large student datasets of **150,000+** records to identify hidden money mule patterns, resulting in actionable insights that reduced potential institutional risk exposure by an estimated **15%**.
- **Strategic Reporting:** Designed 3 custom Power BI dashboards to present actionable risk insights to the CRAIN leadership team, resulting in 2 key institutional risk strategy and policy adjustments.

INDIUM SOFTWARE (Digital Engineering and Enterprise AI Services)
Data & AI Intern

Chennai, Tamil Nadu
06/24 – 09/24

- **AI Product Development:** Engineered a Retrieval Augmented Generation (RAG) prototype utilizing LLMs to process **2,500+** unstructured compliance documents, resulting in the automated extraction of critical business insights that saved 15 hours of manual review per week.
- **Data Engineering:** Built an automated data ingestion framework to clean, validate, and migrate 5 GB of legacy text data into a scalable vector database, resulting in a **40%** acceleration in the retrieval process for complex business queries.
- **Quality Engineering:** Executed rigorous LLM validation and prompt optimization protocols to minimize model hallucinations, resulting in **88%** accurate, deployable application of AI intelligence for 3 cross-functional teams.

ADDITIONAL INFORMATION

- **Technical Skills:** Python, R, SQL, Unix, Relational Database Management Systems (RDBMS), Scikit-learn, TensorFlow, PyTorch, Azure ML
- **Data Visualization & BI:** Power BI, Tableau, Excel, Matplotlib, Seaborn
- **Core Frameworks & Tools:** Git/GitHub, Pandas, NumPy, LangChain, LlamaIndex
- **Certifications:** Google Data Analytics Professional Certificate, Supervised Machine Learning (Stanford University), Entrepreneurship (IIT Madras)
- **Interests:** Running, Football